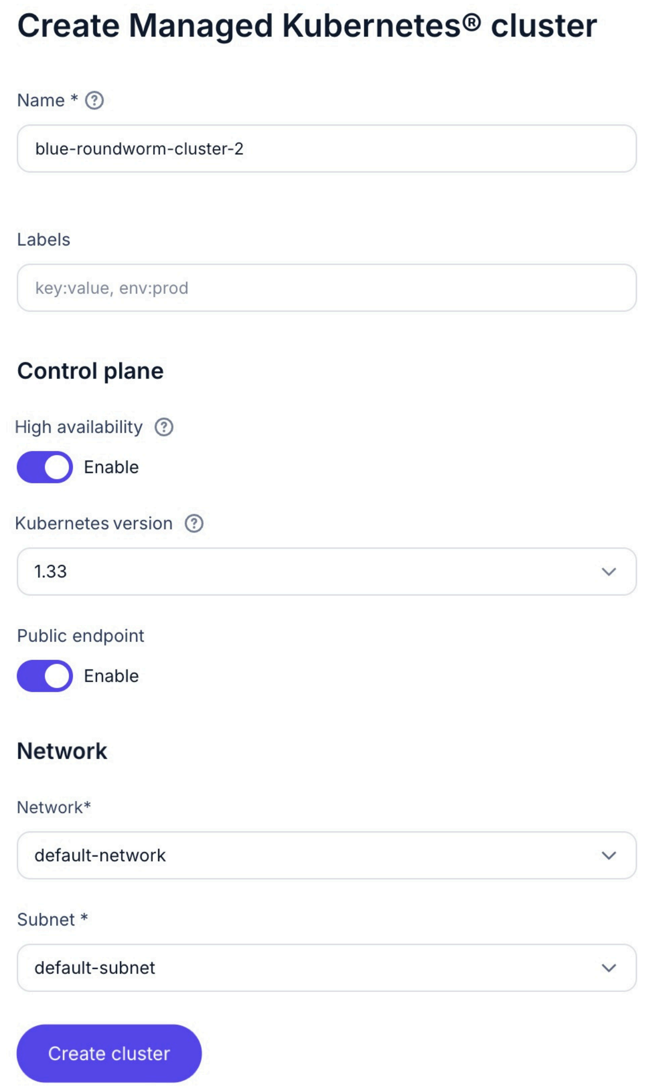
High availability

Under the Control Plane section, the **High availability** toggle switch is enabled by default. At this stage of creating your Managed Kubernetes® cluster, you can set the switch to Disable or Enable.



When this option is enabled, the control plane is deployed across three nodes. Each control-plane node runs an etcd member, forming a highly available distributed etcd cluster.

Note: This option does not affect the cost of the cluster.

Overview and impact

A highly available cluster eliminates single points of failure, ensuring that your system and hosted applications remain operational during hardware or software failures, system updates, or maintenance.

When the option is enabled, the information stored in the etcd database is replicated across multiple control-plane nodes. Any change made to the cluster state is replicated across the etcd members to ensure that each node maintains a consistent copy of the cluster data.

The key idea used here is redundancy: instead of relying on a single component, you have multiple components ready to take over when needed. That makes the system more resilient and keeps it available even when individual machines or components fail. This redundancy helps protect against data loss and minimizes downtime.

If one control-plane node becomes unavailable, the remaining nodes continue to manage the cluster without requiring manual intervention. This allows Kubernetes to continue processing API requests and managing workloads while the failed node is recovered or replaced.

What changes when HA is enabled?

With high availability disabled, your cluster has the following structure:

```
    Worker Nodes

          |

Single Control Plane
-----------------------
API Server
Scheduler
Controller Manager
etcd
```

With high availability enabled:

```
                     Load Balancer

                           |

     --------------------------------------------
     |                    |                       |
     ▼                    ▼                       ▼

Control Plane 1       Control Plane 2       Control Plane 3

API Server            API Server            API Server

Scheduler             Scheduler             Scheduler

Controller Manager    Controller Manager    Controller Manager

etcd                  etcd                      etcd
```

```
        ▲                    ▲                        ▲
        └──────────      Raft Replication      ──────┘
```

Raft is the consensus algorithm used by etcd to replicate data and keep it consistent across the control-plane nodes.

Best Practices

There are multiple benefits to running a multi-node cluster, including improved scalability, better resource utilization, and increased resilience.

High Availability is strongly recommended for production environments. For staging, development, testing, and proof-of-concept environments, it is optional and depends on your operational requirements.

References

For more information on getting started with Managed Kubernetes clusters, see [How to create and modify Managed Service for Kubernetes® clusters].